Staffing Boston's 311 City Service Teams

IE 5374 Applied Generative AI, Module 7 Hackathon Report

Team: Team 8
Members: Ruoxi Chen, Weihe Pan, Hanyue Zhu
GitHub repo: https://github.com/yunxirainbow/boston-311-staffing
Demo video: https://drive.google.com/file/d/1kSjX3XDEYYAYla6GckAmvZLT0EcJZ4Mw/view

Overview

Boston 311 is the city's service request system. In this project we plan how many field crews four city departments need in each month of 2025. The four departments are Public Works, Transportation, Parks and Recreation, and Inspectional Services. We use real 311 data from 2023 and 2024.

Our work has three parts. Part A cleans the data, forecasts monthly demand for 2025 and builds a list of request types. Part B uses the forecast to find the staffing plan with the lowest cost that still closes most requests on time. Part C is our GenAI application. It is a 311 intake assistant that reads a message from a resident and sends it to the right request type and department. It uses retrieval augmented generation, structured output and prompt engineering. Figure 1 shows how the three parts work together.

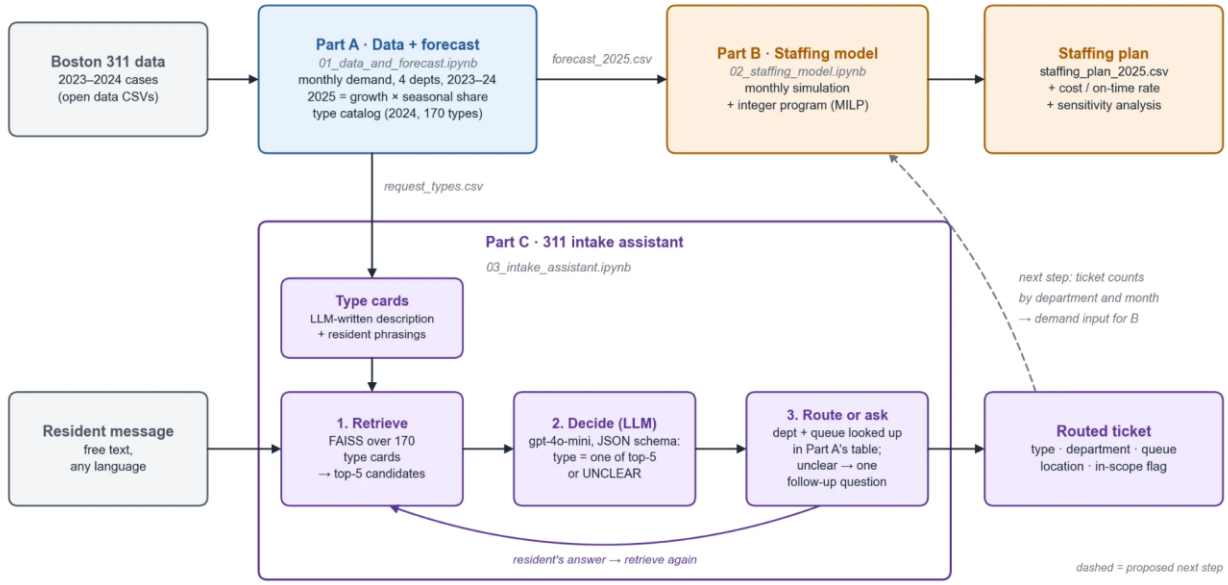


Figure 1. Project architecture. The dashed arrow is a proposed next step. Ticket counts from the assistant could give Part B new demand data every month.

We recommend a plan that closes at least 90% of each month's requests in that month. It costs $14.77M in 2025 and reaches 98% on time. If the city keeps its January crews all year, it would pay $20.07M and only reach 56.7% on time. BTDT is the most understaffed department and should hire early in the year. Parks can cover its summer peak with seasonal crews only.

The intake assistant picks the right request type for 98% of our 45 test messages, and the right department for 98%. When a message could belong to different teams, it asks the resident one follow-up question.

Part A: Data and Demand Forecast

AI use. We wrote the code and the first draft of this part with help from Claude by Anthropic. Then we checked every number against the output of notebook 01_data_and_forecast.ipynb.

1. Data

We used the real Boston 311 request data for 2023 and 2024 from Analyze Boston. The 2024 file has 306,756 requests and the 2023 file has 307,791 requests. We only kept the four departments in scope. They are Public Works, Transportation, Parks and Recreation, and Inspectional Services. In the data their codes are PWDx, BTDT, PARK and ISD. This left 558,663 requests over the two years. Our 2024 count for each department is the same as the number in the topic brief, so we know the data was loaded correctly.

About 16% of requests in 2024 have no target date, but the data still marks them as on time. We tried two ways to compute the on-time rate. The first way uses all requests. The second way only uses requests with a target date. The first way matches the brief for all four departments, with 90% for PWDx, 50% for BTDT, 93% for PARK and 72% for ISD. The second way gives 87% for PWDx, which does not match. So we use all requests when we compute the on-time rate.

2. Demand Patterns

Each department has its own seasonal pattern. PWDx is busiest in August and September. PARK changes the most during the year, and June is about three times January. BTDT and ISD are more stable from month to month.

The trend from 2023 to 2024 is also different for each department. PWDx went down about 8%. BTDT went up about 8%, PARK went up about 7% and ISD went up about 3%. BTDT needs the most attention. Its on-time rate dropped from 65% in 2023 to 50% in 2024, while its requests kept growing. The other three departments stayed about the same, and PARK even went up from 87% to 93%. This shows that BTDT crews could not keep up with the work.

3. 2025 Forecast

We used a simple method that is easy to explain. First, we multiplied the 2024 total of each department by its growth rate from 2023 to 2024. This gives the 2025 total. Then we split the total into months. For each month, we used the average share of that month in 2023 and 2024. This way the forecast keeps both the trend and the seasonal shape. One unusual month in one year also has less effect on the forecast.

The 2025 totals are 145,001 requests for PWDx, 88,044 for BTDT, 20,010 for PARK and 19,145 for ISD. PWDx is lower than 2024 in every month because we assume its downward trend continues. BTDT is higher than 2024 in every month, and September 2025 is above 8,300 requests. These trends may not continue, so Part B tests them in the sensitivity analysis. Figure 2 shows the data for 2023 and 2024 and our forecast for 2025.

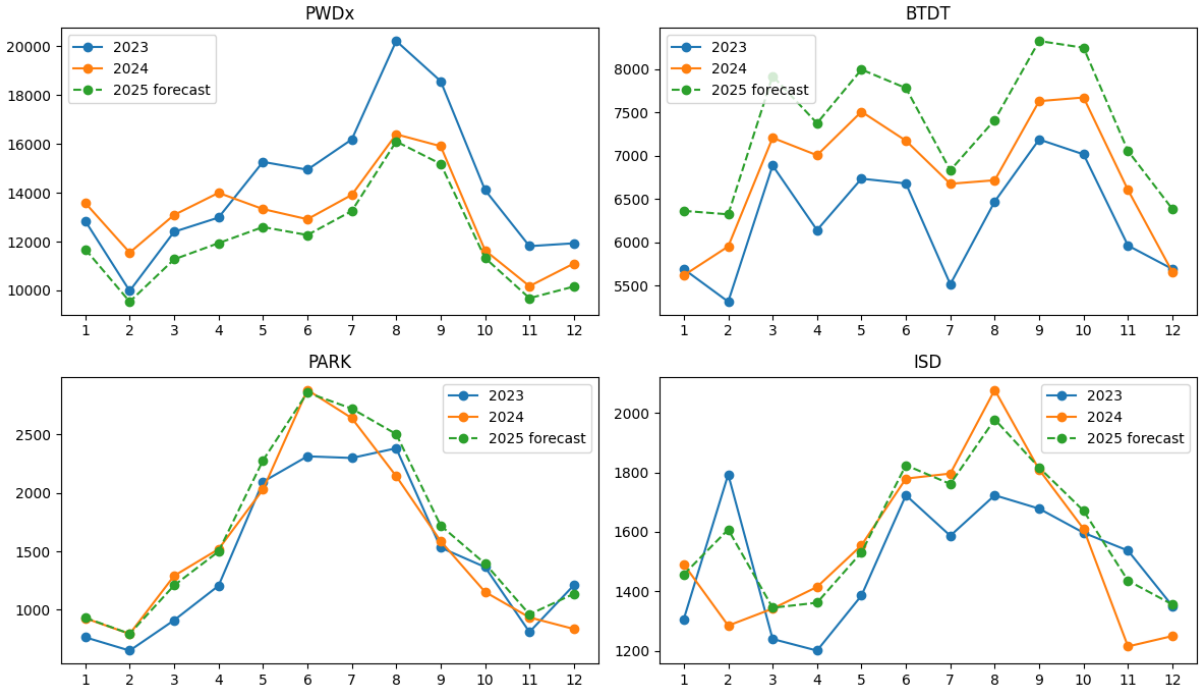


Figure 2. Monthly requests in 2023 and 2024 and the 2025 forecast for each department.

The monthly forecast is saved as forecast_2025.csv. The staffing model in Part B uses it as input.

4. Request Type List

The intake assistant needs a list of request types to search from. We built this list from the 2024 data for all departments. There are 170 request types, which matches the brief. Some types show up under more than one department, so the full list has 589 pairs of type and department. Most of the extra pairs only have a few requests. For each type, we kept the department with the most requests, so each type appears only once. Only 18 types have less than 80% of their requests in this main department. The largest ones are pothole repair, where 75% of requests go to PWDx, and ground maintenance, where 80% go to PARK. The final list also has the most common queue, the reason and a few example titles for each type. It is saved as request_types.csv for the intake assistant in Part C.

5. Other Considerations

Neighborhood equity. On-time rates differ across neighborhoods. In 2024, Charlestown had the lowest rate at 65% and Beacon Hill had the highest at 86%. We also compared neighborhoods inside each department. Here we only counted neighborhoods with at least 200 requests in that department. The gap is largest for BTDT, where West Roxbury is at 37% and South Boston is at 62%. The gap is 15 points for ISD, 14 points for PWDx and 9 points for PARK. Charlestown is also the lowest for PWDx, so the gap is not only because some areas have more parking requests. This means the city should think about where crews work, not only how many crews it has. Some neighborhood names in the data are not consistent, such as Mattapan and Greater Mattapan, so these numbers are only a rough view.

Employee and resident requests. City employees log a large share of requests. The share is 62% for BTDT, 60% for PARK, 50% for PWDx and 40% for ISD. On-time rates are almost the same for employee and resident requests in PWDx and BTDT. In ISD and PARK, employee requests are about 4 to 5 points higher. Crews must handle both kinds of requests, so the staffing model uses the total count. The intake assistant only serves residents, so it covers only about 38% of BTDT requests.

Part B: Staffing Model, Decisions and Sensitivity Analysis

AI use: the model code, charts and first draft of this section were drafted with Claude (Anthropic), then reviewed and checked against the notebook output by the team.

1. Goal

Decide how many field crews each of the four departments (PWDx, BTDT, PARK, ISD) should have in each month of 2025, including when to hire, release, or add Parks seasonal crews, while balancing cost and on-time service.

2. Method

- **Input:** the 2025 monthly demand forecast from Part A (`forecast_2025.csv`).
- **Monthly simulation:** each month, capacity = crews × daily capacity × 21 working days. Requests that can't be finished roll into the next month, and older requests are handled first. Each plan is scored on staffing cost, overdue penalties, and on-time rate.
- **Optimization:** a mixed-integer program (`scipy.optimize.milp`) chooses the number of crews, hires, releases and seasonal crews per month to minimize total cost. We also test versions that require at least 90% or 95% of each month's requests to be closed that month.

3. Assumptions

Given in the brief (used as stated):

- Crew capacity per day: PWDx 12, BTDT 15, PARK 6, ISD 4; 21 working days per month.
- Annual cost per crew: PWDx $130k, BTDT $110k, PARK $120k, ISD $140k.
- January 2025 crews: PWDx 50, BTDT 18, PARK 10, ISD 16. No backlog at the start of 2025.
- New crews need 1 month of hiring and training. Releasing a crew needs 1 month of notice and costs 25% of its annual cost.
- Parks may add seasonal crews from April to October at 1.3× the monthly cost, with no training delay and no release cost.
- Each overdue request costs $20; a request still open 30+ days after its target costs an extra $100.

Our own assumptions:

- January crew levels are fixed; hiring and release decisions take effect from February.
- New crews are paid during their training month.
- A request not closed in the month it arrives misses its target. One still open at the end of the following month is 30+ days late.
- Requests without a target date (about 16%) still use crew time and are treated like other requests.

4. Model check against 2024

We ran 2024 actual demand through the model using January 2025 crew levels. The model predicts lower on-time rates than what really happened (for example, BTDT 29% vs. 50% actual, PWDx 64% vs. 90%). The results would match if about 6–13% of PWDx, BTDT and ISD requests needed no crew visit (for example, duplicates or requests closed without field work). Parks matches only at about 55% because many Parks requests have 365-day targets. This means our crew numbers are **conservative (an upper bound)**.

5. Key decisions

Plan comparison (all four departments, 2025)

Plan	Total cost	Overdue penalties	On time
Keep January crews all year	$20.07M	$8.15M	56.7%
Lowest cost (no service target)	$14.12M	$0.70M	87.2%
90% service target (recommended)	**$14.77M**	**$0.11M**	**98.0%**
95% service target	$14.95M	$0.06M	98.8%

- **Keeping January crews is the most expensive option.** BTDT, PARK and ISD fall behind, and overdue penalties build up to $8.15M.
- **The lowest-cost plan accepts many late requests.** With the brief's numbers, a crew can cost more per request than a late request does (ISD costs about $139 of crew time per request), so this plan leaves ISD at 58% and Parks at 75% on time.
- **We recommend the 90% target.** It costs $0.65M more than the lowest-cost plan and raises on-time service to 98%. Moving to 95% adds another $0.18M for less than one extra point.

Recommended plan by department

Dept	Jan	Peak	Dec	Changes	Cost	On time
PWDx	50	60 (Aug)	51	Add 2 crews from July and 8 from August; release 1 in September and 8 in October	$7.22M	98.1%
BTDT	18	26	26	Add 4 crews from February, 2 from March, 1 from April, 1 from September	$2.79M	98.0%
PARK	10	10 + 12 seasonal	10	No full-time hires; seasonal crews Apr–Sep: 2, 8, 12, 12, 10, 4	$1.83M	98.4%
ISD	16	22	22	Add 3 crews from February, 2 from June, 1 from August	$2.93M	96.4%

- **BTDT is the most understaffed department.** It needs about 23 crews on average but has 18, so it hires early in the year.
- **Parks uses only seasonal crews for its summer peak.** Paying 1.3× for a few months is cheaper than full-time crews plus release costs.
- **PWDx only needs extra crews for the August–September peak.** Releasing them in October costs about the same as keeping them until December, so the city could keep them if it expects the same peak in 2026.

6. Sensitivity analysis

We kept the recommended plan fixed and tested it under different demand.

Scenario	On time	Total cost
Base forecast	98.0%	$14.77M
Demand −20%	100%	$14.66M
Demand +20% all year	34.5%	$23.20M
Snowstorm: February +50% for PWDx and BTDT	88.9% (BTDT 72%)	$15.28M
Heat wave: July +40% for PARK and ISD	94.7% (PARK 70%)	$14.95M
Lighter workload (only 90% of requests need crews)	100%	$14.66M

- The plan uses crews at 92–96% of capacity, so it has little extra room.
- **A one-month spike** (snowstorm or heat wave) is cleared within one or two months.
- **A sustained 20% increase** causes backlogs to grow and costs to rise sharply. If this increase were known in advance, re-optimizing would bring on-time back to about 97% at $17.7M instead of $23.2M.

- **Recommendation:** re-run the model every month with actual demand. Because new crews take one month to start, the city should hire as soon as on-time drops toward 90%. Parks seasonal crews are the fastest option because they need no training.

7. Limitations

- The model works by month, so it doesn’t capture timing within a month.
- Real target dates differ by request type; we treat any request not closed in its arrival month as late.
- The brief’s crew capacity seems to overstate the real workload (see Section 4).
- The plan ends in December 2025 and doesn’t plan for the 2026 peak.

Part C: 311 Intake Assistant, Design and Evaluation

AI use: the code, the first draft of the test set and the first draft of this section were written with Claude (Anthropic), then reviewed and checked against the notebook output by the team. The type descriptions used for retrieval are generated by gpt-4o-mini inside the notebook.

1. Goal

Turn a resident's free-text message into a 311 ticket for the right team. The ticket has four parts: the request type (one of the 170 real 2024 types), the department, the queue and the location. When a message could belong to different teams, the assistant asks one follow-up question instead of guessing. This is the project's GenAI application. It uses three course techniques: retrieval-augmented generation (RAG), structured output and prompt engineering.

2. Method

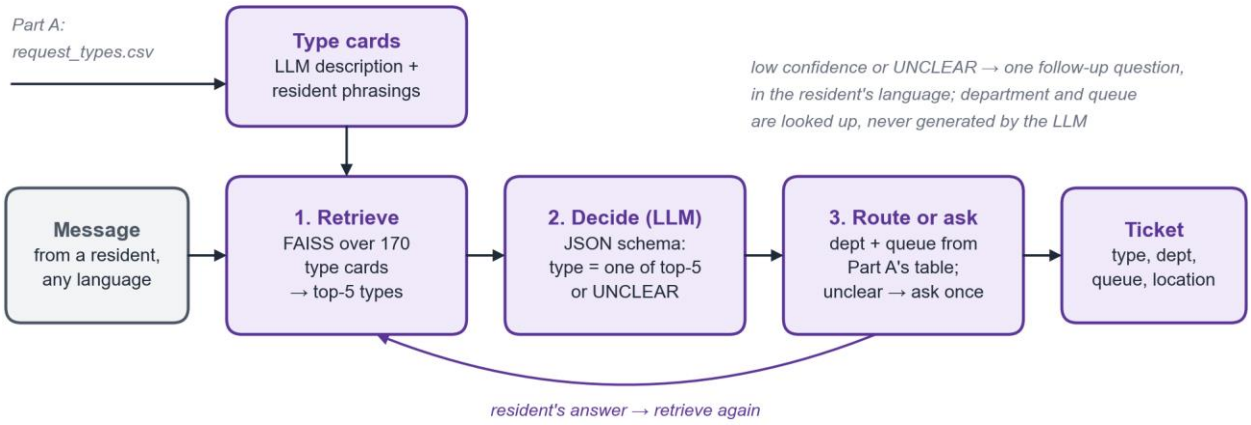


Figure 3. Intake pipeline (notebook 03_intake_assistant.ipynb).

- **Catalog:** Part A's request_types.csv lists 170 request types. Each type is assigned to the department that handles most of its cases, together with its most common queue and example case titles.
- **Type cards (LLM data enrichment):** for 91 of the 170 types, the example titles only repeat the type name (for example "Parking Enforcement"). For each type, gpt-4o-mini writes a 1–2 sentence description and 4 short messages a resident might send. It uses a strict JSON schema. The cards are cached, so this step runs once.
- **Retrieval:** each card is embedded with text-embedding-3-small and indexed with FAISS. The 5 closest types become the candidates. After a follow-up, retrieval runs twice: once on the answer alone and once on the message plus the answer. The two lists are merged, up to 10 types.
- **Decision:** gpt-4o-mini, at temperature 0, returns JSON with the request type, a confidence level (high, medium or low), the location and an optional follow-up question. The JSON schema is built for each call, so the type must be one of the candidates or UNCLEAR.
- **Routing:** the department, the queue and an in-scope flag are looked up in Part A's table. They are never generated by the LLM. A type is in scope if its department is PWDx, BTDT, PARK or ISD.
- **Follow-up rule:** the assistant asks one question when confidence is low or the type is UNCLEAR. Confidence is low when two or more candidates handled by different teams fit about equally well. The assistant asks at most once per message.
- **Demo:** a Gradio chat app in the notebook. If the assistant asks a question, the resident's next message is treated as the answer.

3. Design decisions and assumptions

- **Retrieve first, then let the LLM choose.** Retrieval keeps the prompt short: 5 candidates instead of 170. We test the alternative, putting all 170 types in the prompt, as set-up E in Section 4.
- **Constrain the output.** The schema only allows real candidate names, so the model cannot invent a request type. The department and queue come from data. A made-up type would have no team to go to.
- **Keep all 170 types.** 22 types belong to teams outside the four planned departments. Examples are Boston Water and Sewer (hydrants, catch basins) and the needle program (10,710 requests in 2024). We keep these types and mark them as out of scope, instead of forcing them into one of the four departments.
- **Ask only when the answer changes the team.** A question costs the resident time, so "low confidence" means the candidates belong to different teams. For example, "My landlord won't fix anything" fits three housing types, but all three go to the same ISD team, so the assistant files it without asking.
- **Answer in the resident's language.** The follow-up question is written in the language the resident used, and the test set includes Chinese and Spanish messages. This supports equal access for residents who do not write in English, one of the equity questions in the brief.
- **What we score.** We score the request type and the department. The queue is the type's most common queue (top_queue). Some Public Works queues are split by district, and routing to a district would need a lookup from location to district.
- **Assumptions.** The 2024 type catalog and department assignments still apply in 2025. Each type goes to its main department, as in Part A. The test messages are written the way residents write.

4. Evaluation

We wrote 45 test messages (intake_test_set.csv, version 2) with AI help and checked every label by hand against the type list. Some messages accept 2–3 equally valid types. The clear messages include typos, lower-case text, Chinese and Spanish.

Category	n	Counted as correct when
Clear	30	the predicted type is one of the acceptable types
Ambiguous	9	the assistant asks, and after the scripted answer the type is acceptable
Other department	5	the right type outside the four departments, flagged as out of scope
Not a 311 request	1	the assistant returns UNCLEAR and asks instead of forcing a type

Table 1. Test set (45 messages).

Scoring is end to end. If the assistant asks a follow-up and the row has a scripted answer, we score the answer given after the follow-up. Three ambiguous messages appear twice, with answers that point to different teams. For example, "The light on my corner is broken" can mean a street light (Public Works) or a traffic signal (Transportation). An assistant that guesses can get at most one message in each pair right. We compare five set-ups to show what each part adds.

Set-up	Type acc. (all)	Type acc. (clear)	Dept. acc.	Retrieval recall	Asks when it should	Asks when it shouldn't
A Embedding top-1, basic cards	49%	53%	80%	80%	0%	0%
B Embedding top-1, type cards	60%	70%	82%	89%	0%	0%
C RAG + LLM, basic cards	76%	87%	89%	82%	40%	0%
D RAG + LLM, type cards (final)	**98%**	**100%**	**98%**	**98%**	**70%**	**0%**
E All 170 types + LLM	89%	100%	98%	n/a	10%	0%

Table 2. Results on the 45-message test set. RAG = top-5 retrieved candidates; E puts all 170 types in the prompt instead. Basic cards contain only Part A's fields. Retrieval recall is the share of messages with an acceptable type among the candidates. "Asks when it should" is measured on the 10 messages that need a question (9 ambiguous, 1 not a 311 request).

- **Type cards give the biggest gain.** Without the LLM, they raise type accuracy from 49% to 60% (A → B). With the LLM, they raise it from 76% to 98% (C → D). Retrieval recall rises from 80–82% to 89–98%.
- **The LLM step matters.** With the same type cards, letting the LLM choose among the top 5 raises accuracy from 60% to 98% (B → D). The embedding baselines can never ask a question.
- **Retrieval beats the full catalog on ambiguous messages.** D and E are equal on clear messages (100%). With all 170 types in the prompt, the model asked in only 1 of the 10 cases that needed a question; D asked in 7. As a result, E guessed wrong on the ambiguous pairs (89% vs. 98% overall). D's prompt is also much shorter.
- **No unnecessary questions.** No set-up asked a follow-up on a message that did not need one.

5. Error analysis (set-up D)

Only three of the 45 messages did not get the expected result:

- **T42, "The light on my corner is broken" (the resident meant the traffic light).** This is the only wrong type. All five retrieved candidates were lighting types such as Street Light Outages and Street Light Knock Downs, and neither traffic-signal type was among them. With no Transportation option to compare, the model was confident and did not ask. A possible fix is department-diverse retrieval: when the top results all come from one team, add the best match from another team. We did not test this, to avoid tuning on the test set (see Section 6).
- **T30, the same message (street light).** The assistant did not ask here either, but its guess was right.
- **T43, "There's trash everywhere on my street".** This is the same message as T31. On T31 the model rated its confidence low and asked. On T43, in a separate call, it did not rate its confidence low and answered directly. Its guess, Illegal Dumping, happened to be right. Even at temperature 0, the API can return a different confidence for the same message when it is close to the medium/low boundary.

6. Test-set revisions (v1 → v2)

We ran the first version of the test set (v1, 41 messages) before any prompt work, then reviewed the errors of set-up D. We fixed one bug and the test set. We did not change the system prompt.

- **Bug fix:** after a follow-up, retrieval used the message and the answer as one query. The vague message drowned out the answer, so the right type dropped out of the candidates. Retrieval now also runs on the answer alone.

- **Label fixes:** one message gained a second acceptable type with the same department and queue. Another moved from ambiguous to clear, because all its candidates go to the same ISD team.
- **Harder ambiguous cases:** in v1, every scripted answer pointed to the most common reading, so guessing scored as well as asking. Set-up E scored 98% while asking only 14% of the time. v2 repeats three messages with answers that point to the less common reading, and adds a new ISD case.

On v1, set-up D scored 95% type accuracy, 100% department accuracy, and asked in 71% of the cases that needed a question. Because v2 was written after we saw the v1 errors, the v2 numbers are not a blind test.

7. Limitations and next steps

- **Small, non-blind test set.** The 45 messages were drafted with AI and revised after v1. A fairer check would be 10–20 messages written by teammates who have not seen the system. Any further change, such as department-diverse retrieval, should be tested on such a set.
- **Resident vs. staff wording.** 42% of 2024 requests were logged by city employees, not residents. Staff reports may use different wording and internal type names. Our test set only covers resident-style messages, so it does not measure this.
- **Retrieval can hide the alternative (T42), and confidence varies from run to run near the boundary (T31 vs. T43).** Both can make the assistant skip a question it should ask.
- **Queue routing.** We route to the type's most common queue. Public Works queues that are split by district need a lookup from location to district.
- **Link to Part B.** Classifying incoming messages and counting tickets by department and month would give Part B live demand data, so B can re-run its plan every month as its sensitivity analysis recommends. This is the dashed arrow in the project architecture diagram.

AI Use

We used AI tools in two ways. First, our product uses OpenAI models. The intake assistant uses gpt-4o-mini to write type descriptions and to pick the request type. It uses text-embedding-3-small for retrieval. Second, we used Claude by Anthropic to help us write code and the first drafts of this report. Each part starts with a short note that says which pieces were drafted with AI. We reviewed all AI output and checked every number in this report against our notebook output.